

ENV 6128

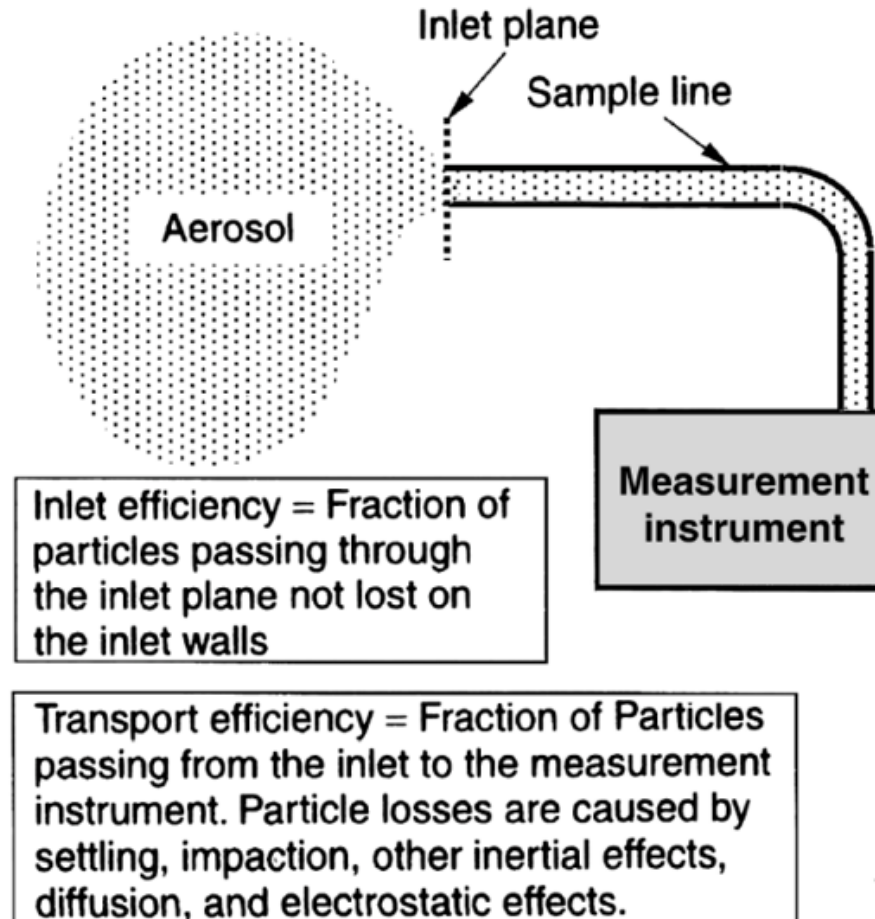
Smart Air Quality Monitoring & Air Pollution Control

Today's Topics

- Aerosol sampling and CPC
- Expectation
 - Understand aerosol sampling loss, and considerations for ambient aerosol sampling
 - Understand the working principles of different CPC technologies for measuring aerosol number concentrations

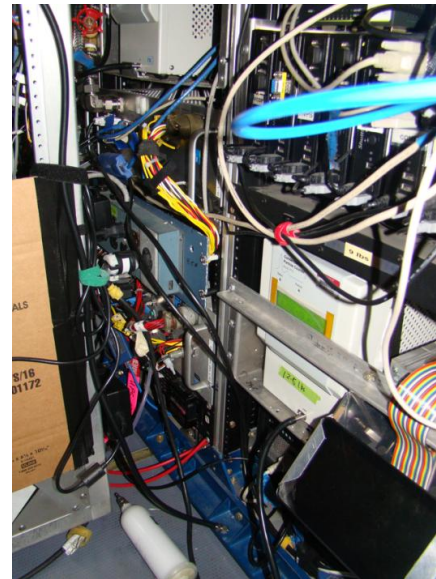
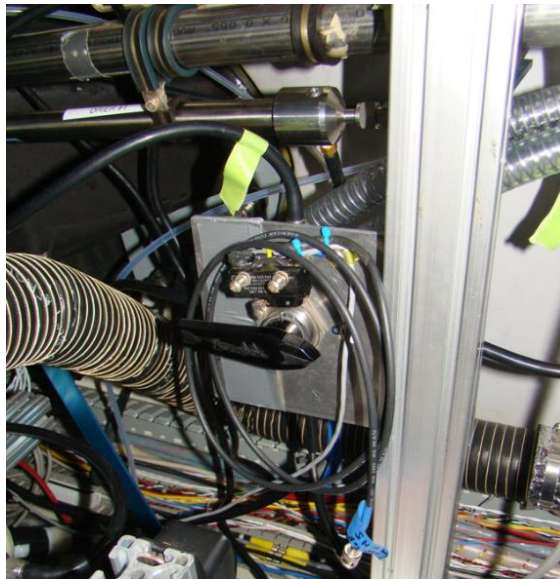
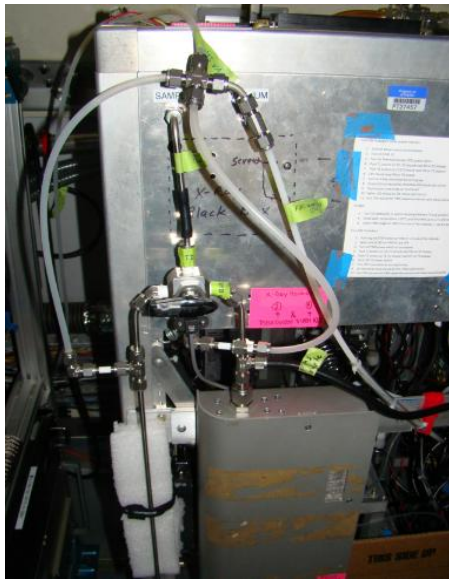
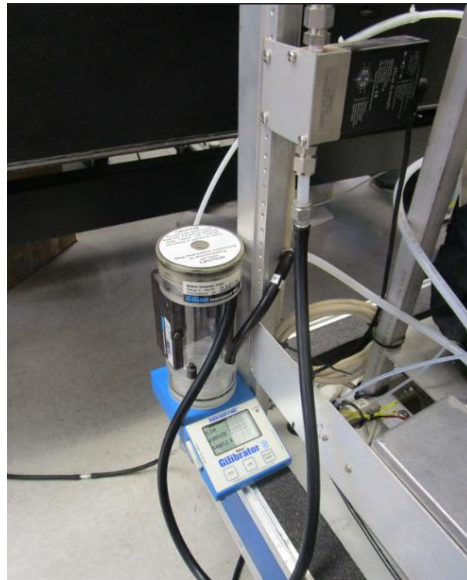
Aerosol Sampling

Sampling efficiency



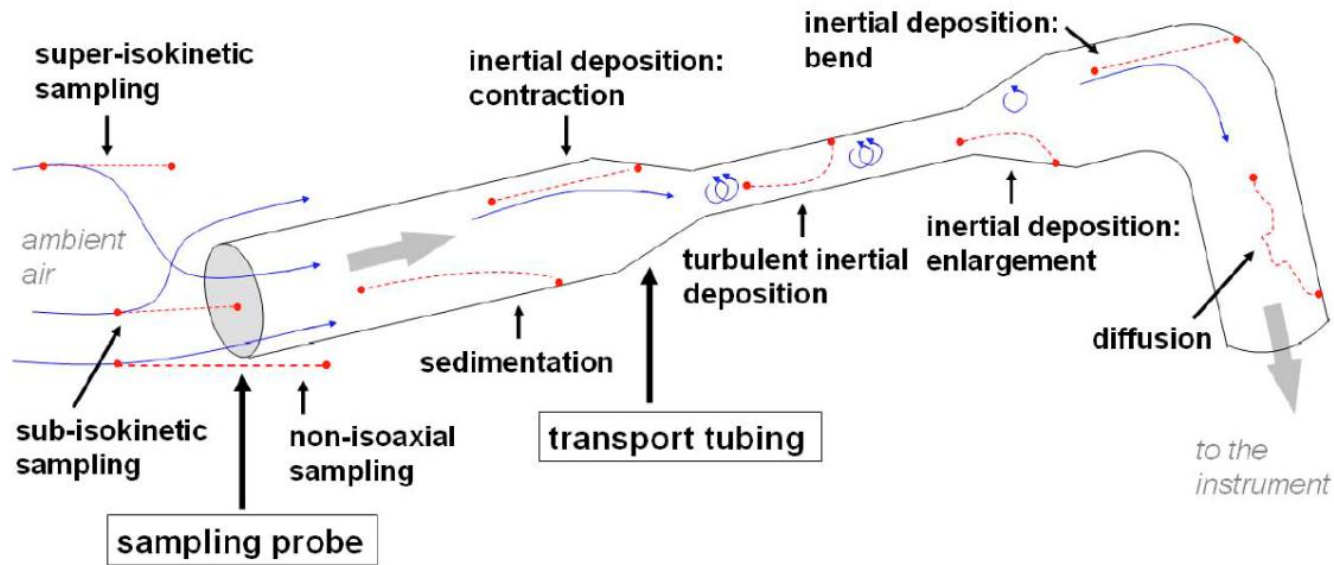


Example aircraft sampling system





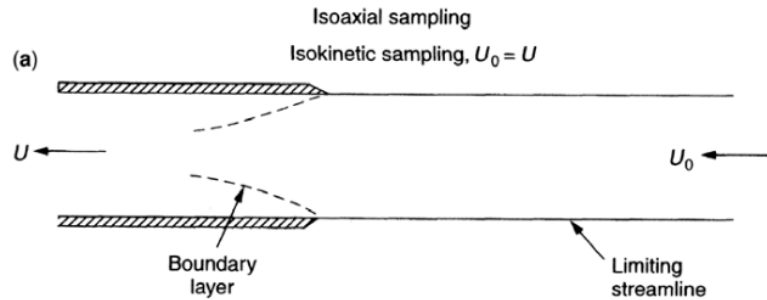
Aerosol sampling loss



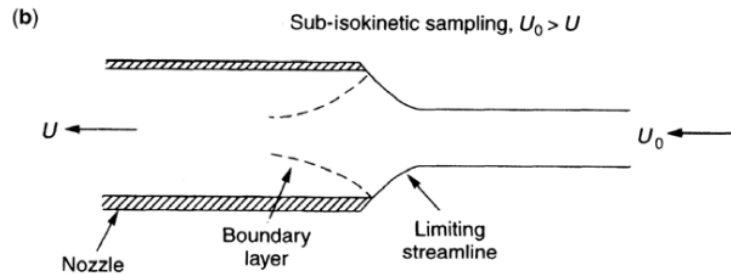
1. Gravitational settling.
2. Diffusional deposition.
3. Turbulent inertial deposition.
4. Inertial deposition at a bend.
5. Inertial deposition at flow constrictions.
6. Electrostatic deposition.
7. Thermophoretic deposition.
8. Diffusiophoretic deposition.

Isoaxial sampling

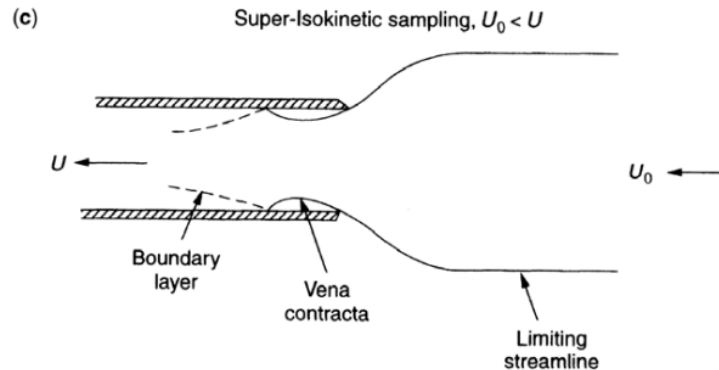
A inlet facing the gas flow where the direction of the sample flow is aligned with that of the gas flow.



Isokinetic

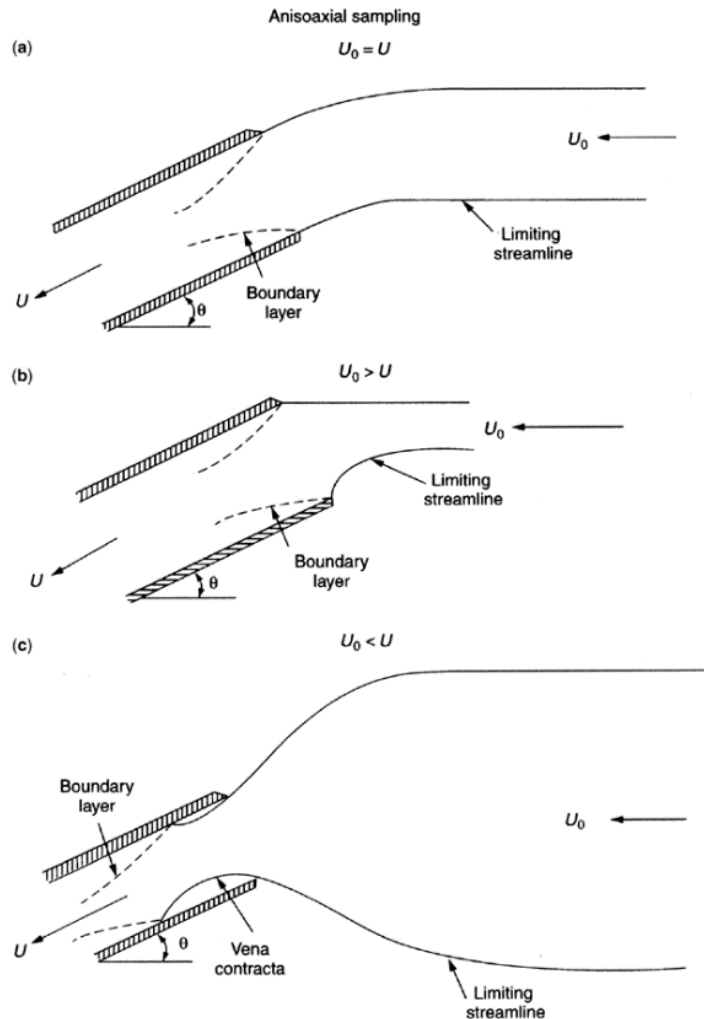


Sub-Isokinetic
Over-sample large aerosol



Super-Isokinetic
Under-sample large aerosol

Anisoaxial sampling



- Additional transmission loss due to particle impaction on inside lip of nozzle

Importance of Calibration

- Sampling systems should be calibrated fully assembled for aerosol sampling and transport efficiency.
- Calibration should be done at the gas flows and over the size range of interest.
- Calibration of component sections may be sufficient.
- Calibration at other than operation conditions, in combination with models and correlations.
- Sampling efficiency can be estimated if the system consists of components with well-characterized data and models available in literature.

Sampling in calm air

Condition for representative sampling

$$Stk_i \leq 0.016$$

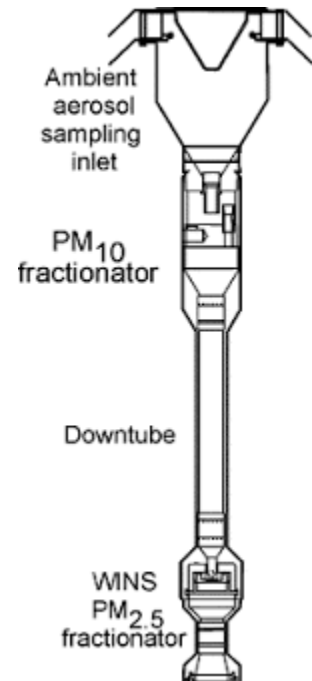
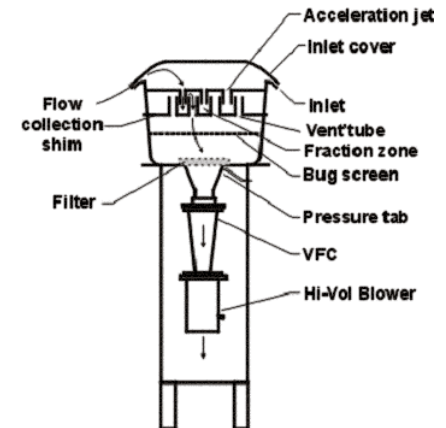
$$\frac{V_{ts}}{U} \leq 0.04$$

Davies (1968)

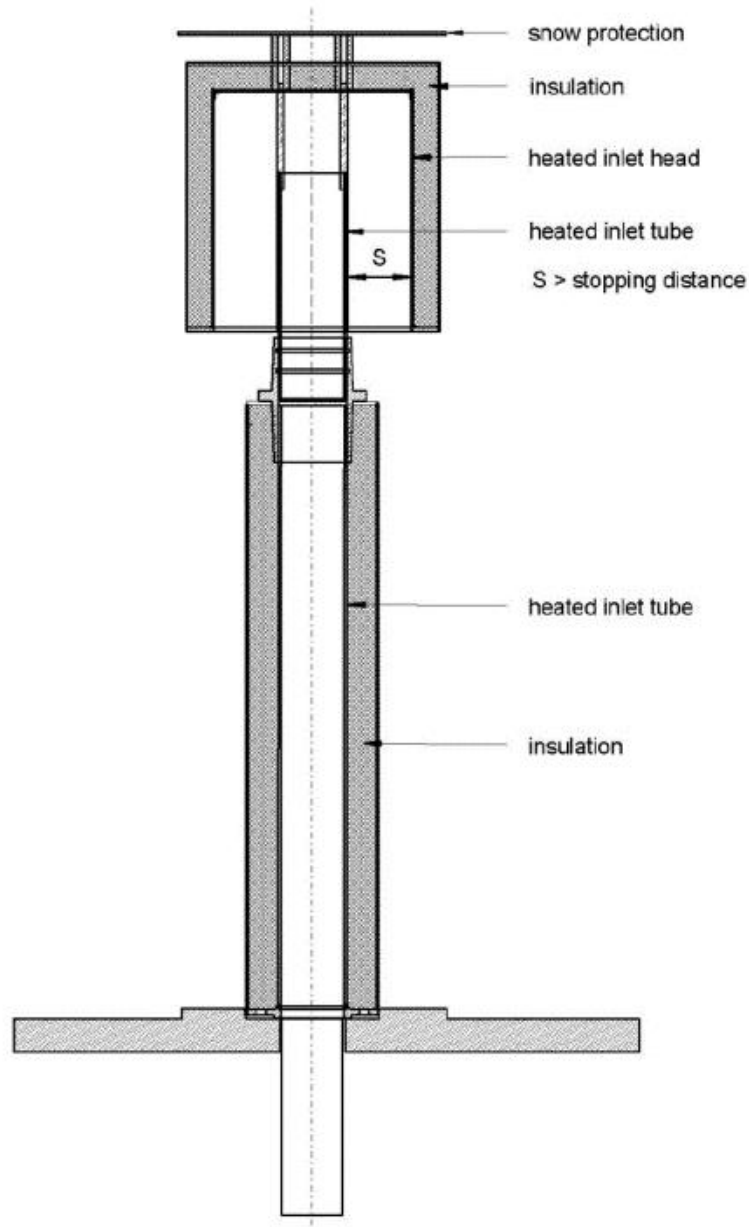
Upward facing inlet

$$Stk_i \frac{V_{ts}}{U} = \frac{\tau V_{ts}}{d} \leq 0.05$$

(Agarwal and Liu 1980)



Schematic of a whole air inlet

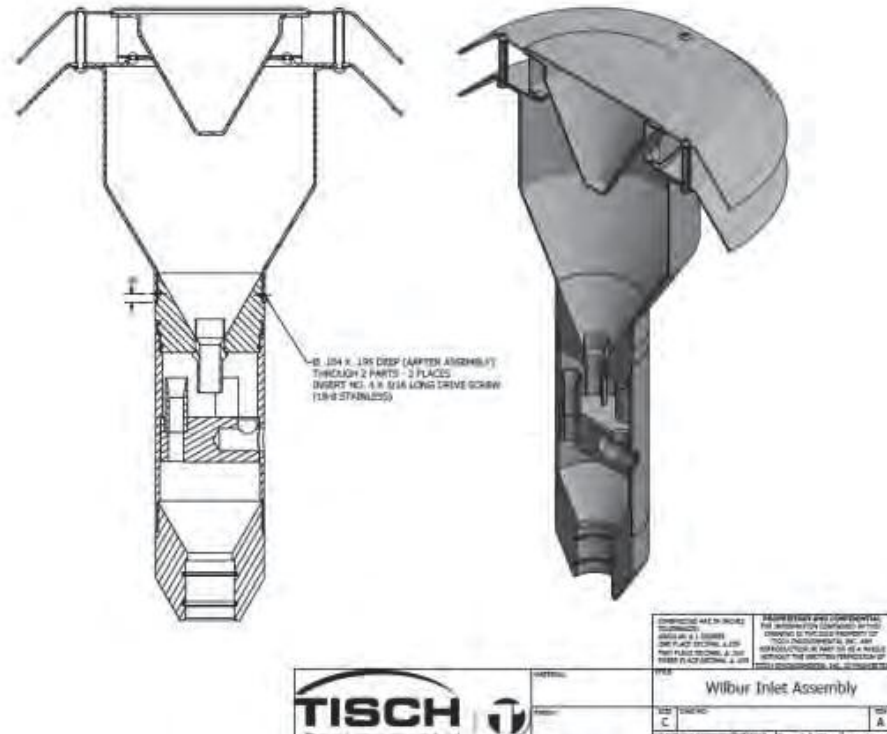


As used in World
Meteorological
Organization (WMO)
Global Atmosphere
Watch (GAW) network

PM₁₀ sampling inlet



Tisch TE-PM10-D



- Low-volume inlet
- Rate at 16.67 Lpm
- Winter tunnel tested

Ideal aerosol sampling system

- Excludes precipitation from the sampled aerosol
- Provides a representative ambient aerosol sample with minimal diffusional and inertial losses
- Provides aerosol particles at a low relative humidity ($< 40\%$)
- Minimizes the evaporation of volatile particulate species

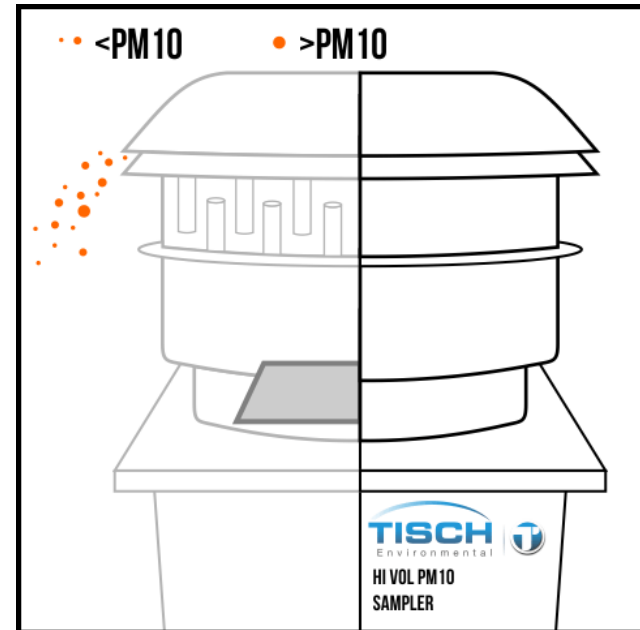
Inlet design considerations

- Sample air should be brought into the laboratory through a vertical stack.
- Inlet that is high enough above ground level to minimize local influences (a minimum height of 2 m above the roof, even higher inlets may be required for obstructed sites).
- If possible, the sampling duct should be brought through the roof of the laboratory. If this is not possible, then a gradual bend in the inlet pipe is acceptable.
- High aerosol transmission efficiency that does not vary with wind direction or wind speed. This can be achieved with a vertical air duct with a cover that excludes drizzle, rain and snow mounted on top of the inlet duct.
- The inlet should have a particle cut-off diameter corresponding to sampling goal under ambient conditions.

Tubing and flow splitters

- For the sampling of particles, the pipes should be made of conductive material, and electrically grounded.
- Short pieces of tubing might be replaced by conductive silicone tubing, which is elastic and conducting at the same time.
- A perfect inlet installation also avoids sources of turbulence (bends, connectors) as best as possible (turbulence enhances particle losses due to diffusion) and keeps the sampling lines as short as possible.
- Isokinetic flow splitter: a sample is removed from the core of the main aerosol flow rather than from streamlines near the wall of the main pipe. This ensures a representative sampling especially of coarse and nano-particles.

High volume inlet



TISCH animation

<https://tisch-env.com/high-volume-air-sampler/pm10>

TISCH High Volume Sampler TE-6070V

Sample drying

- Dry sampling maintains the RH at a sufficiently low range that hygroscopic growth effects are minimized.
- The preferred methods for reducing the relative humidity are to remove water vapor with a diffusion or membrane dryer, alternatively to reduce the dewpoint temperature by diluting the sample with filtered, dry air.
- An alternative method is to reduce the RH to 40% by modest heating to a temperature not to exceed 40°C (to minimize loss of semi-volatile species).

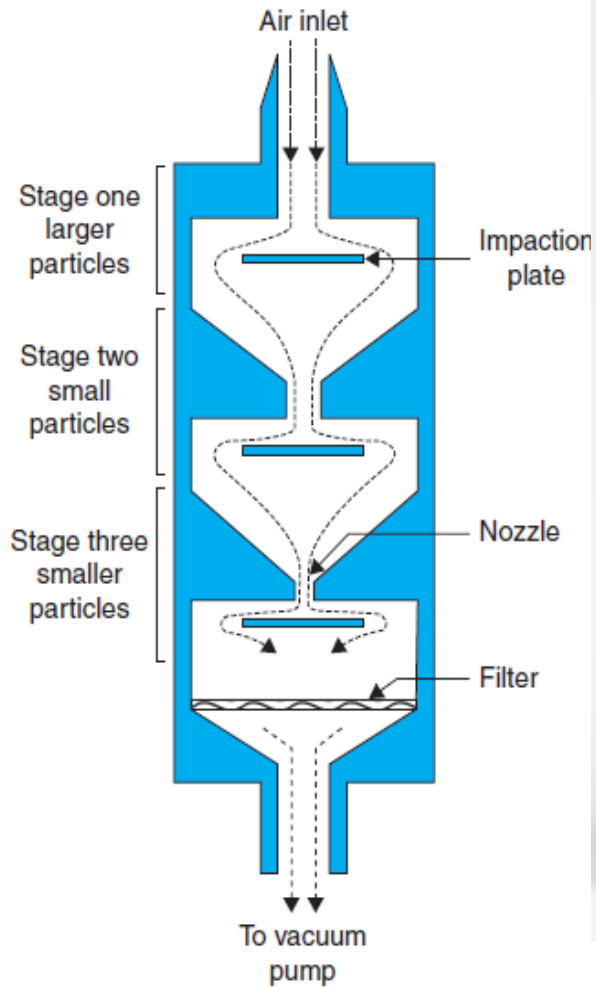


Thermo Scientific 1405 TEOM
Note heated inlet

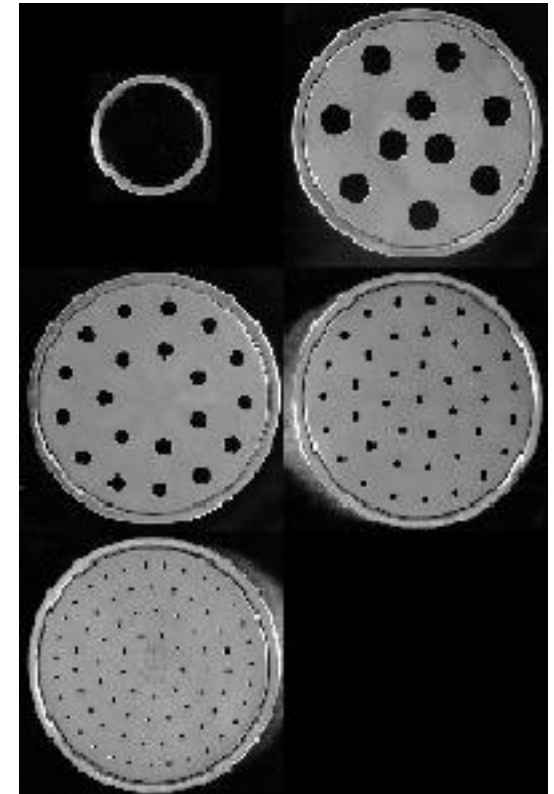


Teledyne T640
Note heated tube

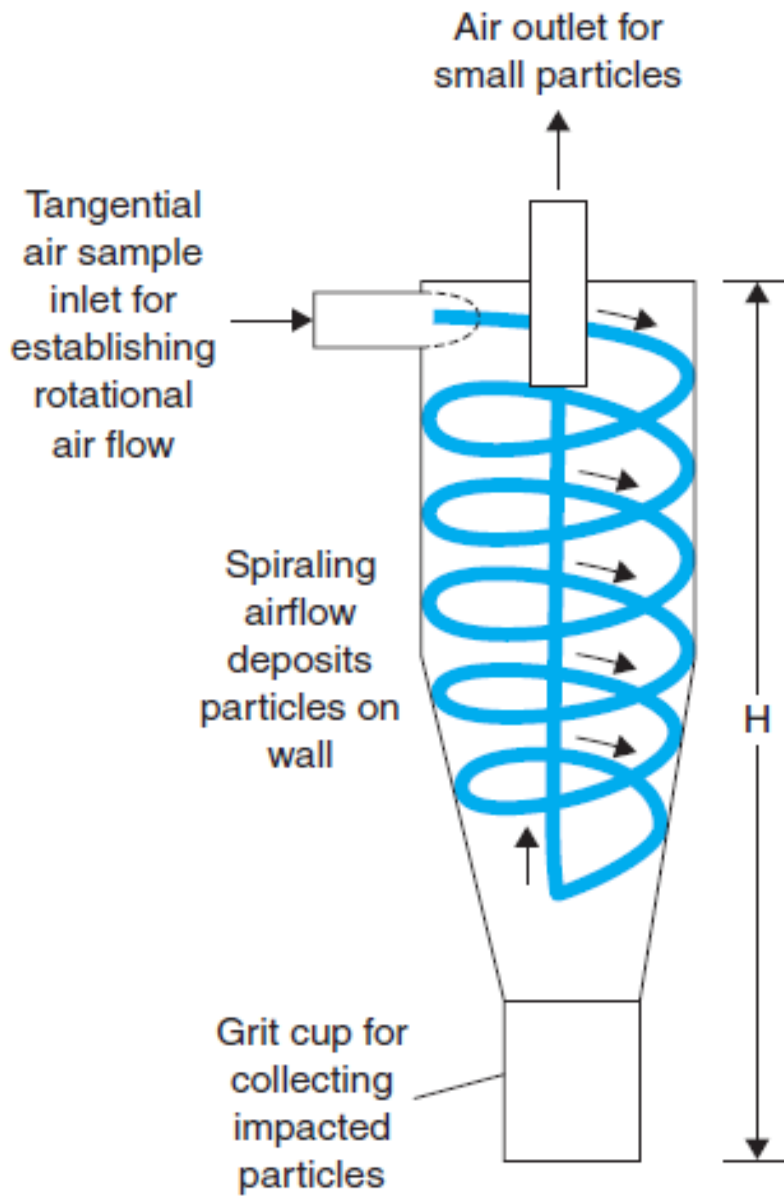
Cascade impactor



(A)



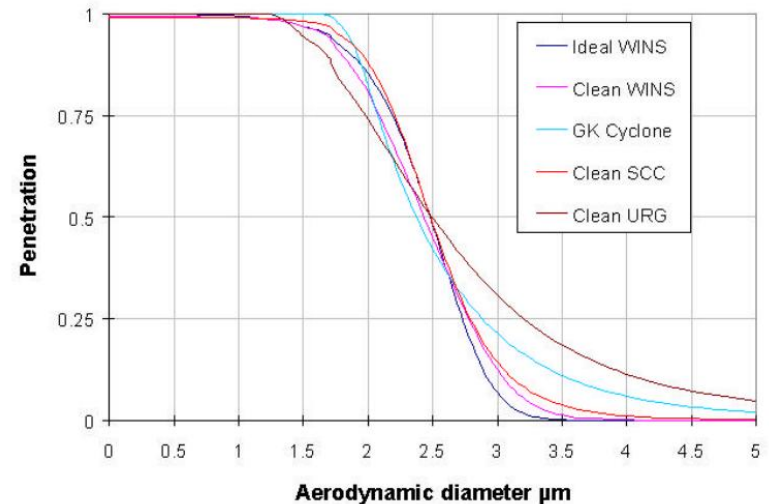
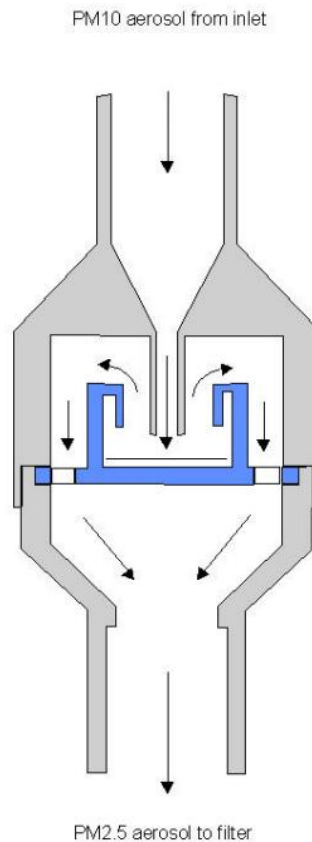
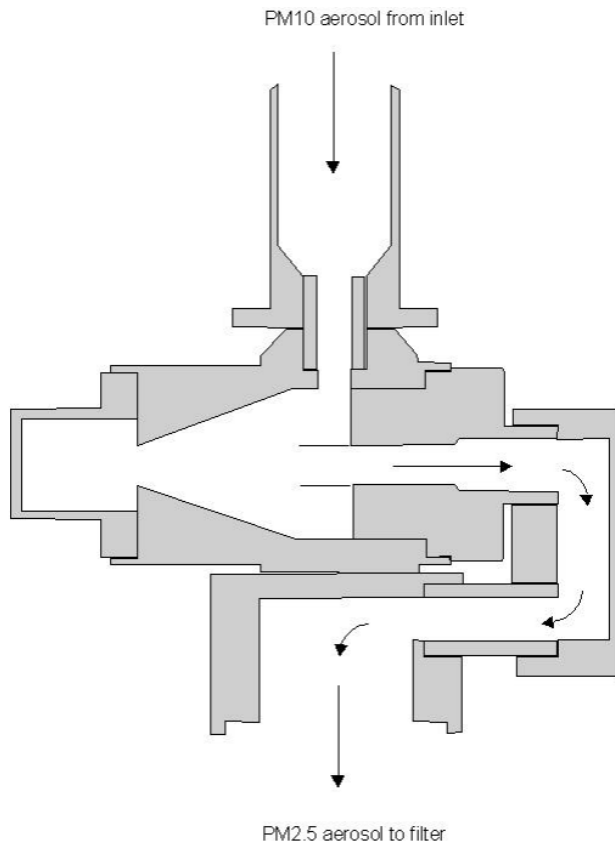
Cyclone pre-collector



Source: The University of California Air Pollution Health Effects Laboratory, with kind permission.

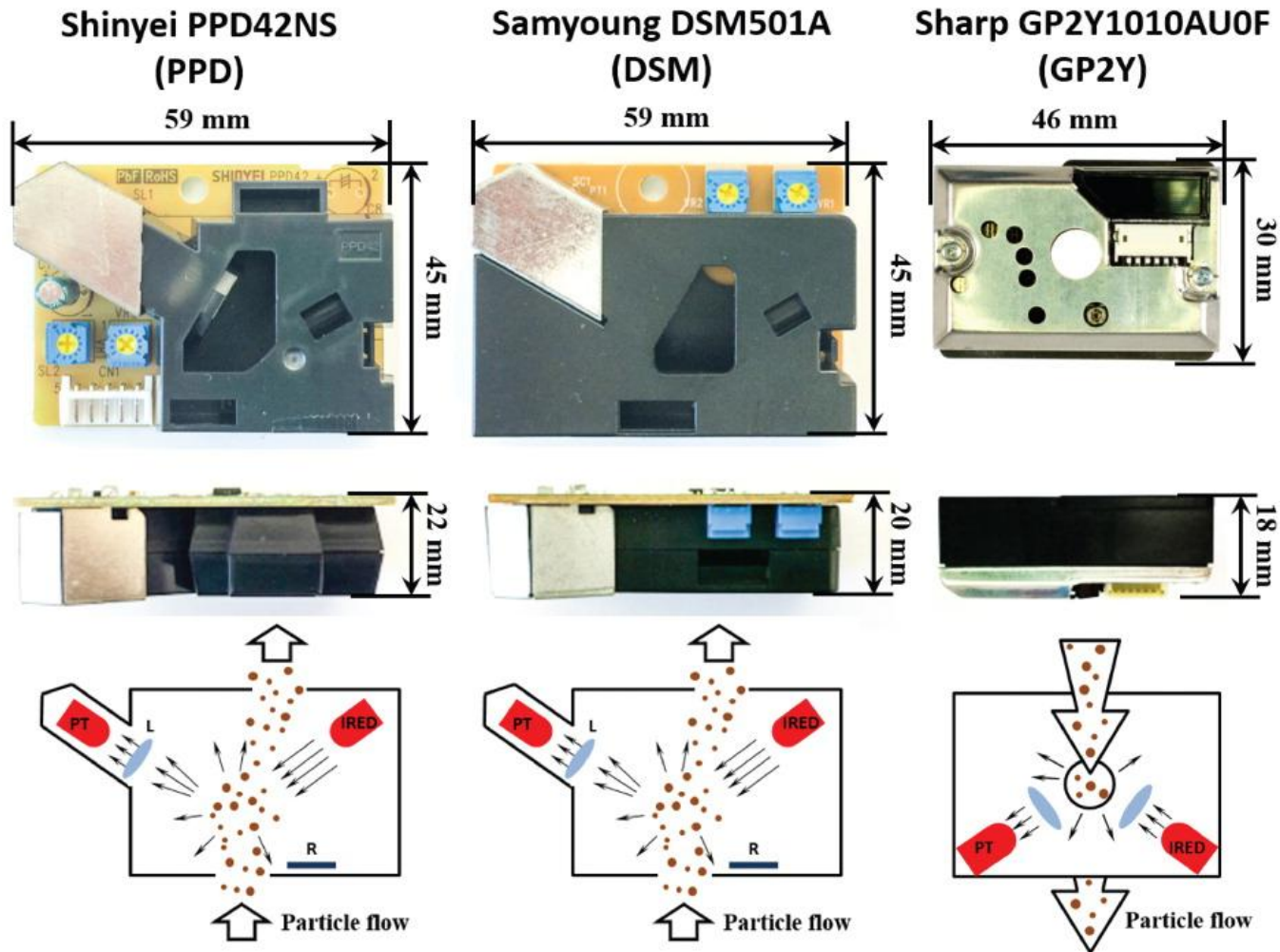
Cyclone & impactor

- Separate particles of certain size at certain flow rate
- Also used in industrial hygiene for personal sampling



Penetration curve of SCC
& WINS

Sampling issues in low-cost sensor



Yang Wang et al: <https://doi.org/10.1080/02786826.2015.1100710>

Plantower PMSA003: <http://www.plantower.com/content/?125.html>

Sensirion SPS30: <https://www.mistywest.com/posts/teardown-sensirion-particle-matter-sensor/>

Condensation Particle Counter

Particle counting

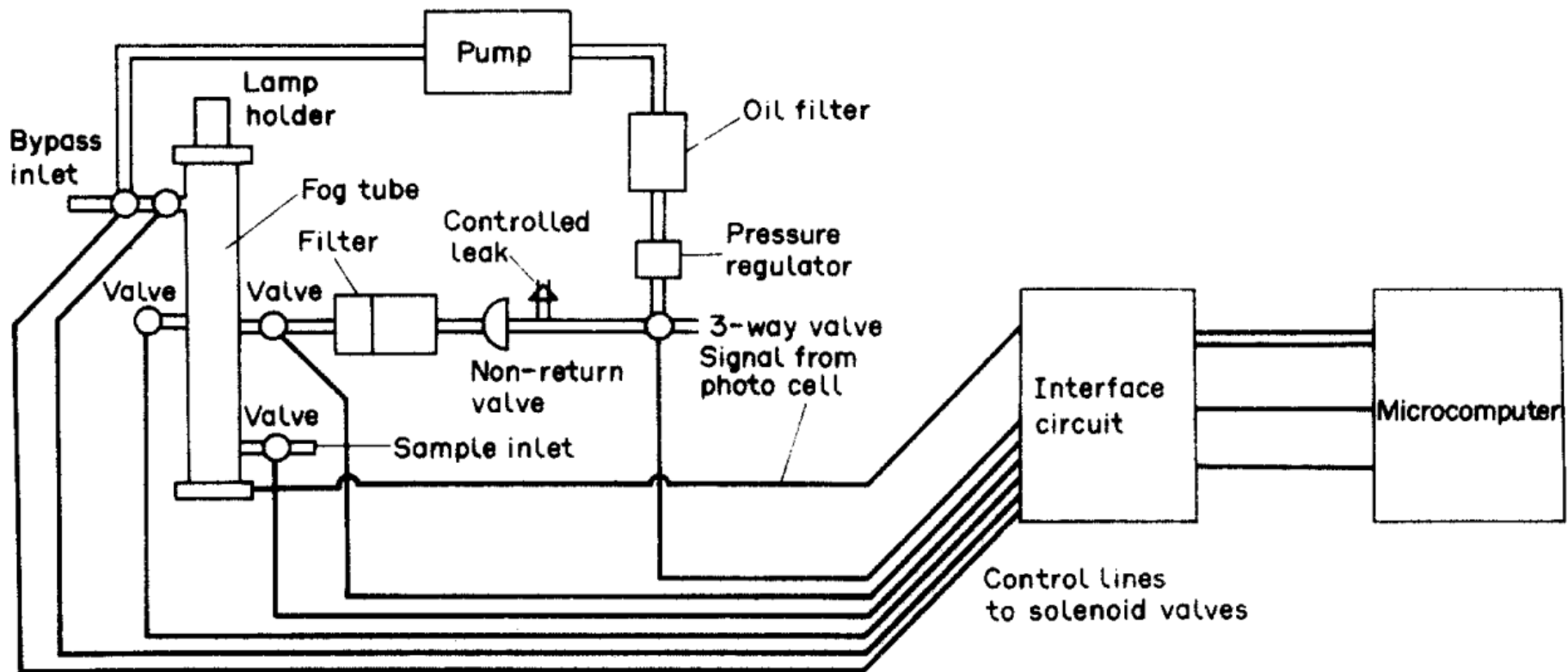
- Light scattering
 - $50 \text{ nm} < D_p < 5\text{-}100 \text{ }\mu\text{m}$
- Condensation Particle Counter (CPC)
 - Generation of a supersaturated condition
 - Condensation growth of particles
 - Detection of grown particles by light scattering
- Performance
 - Detection efficiency (activation efficiency, transmission efficiency)
 - Response time

Supersaturation

- Cloud in a bottle
 - <https://www.youtube.com/watch?v=WeXuKd0vMRk>
- Supersaturation: $S = p/p_s > 1$
 - p : partial pressure
 - p_s : (saturation) vapor pressure
- Antoine equation
 - How p_s change with temperature
 - A , B & C are constants

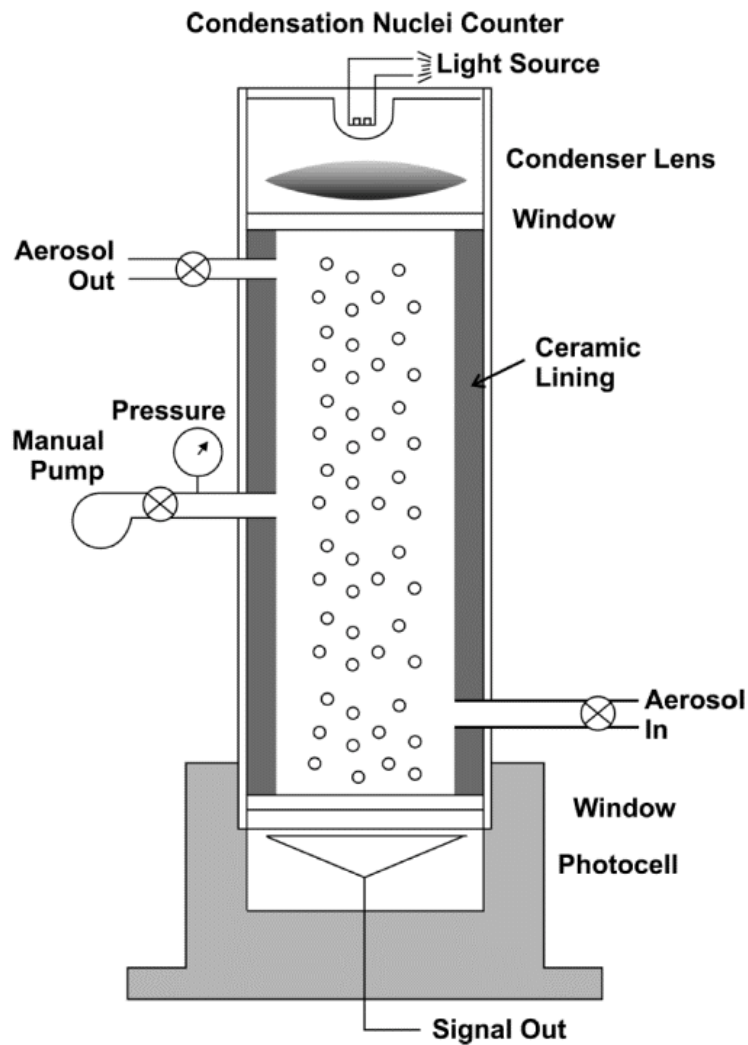
$$\log_{10}(p_s) = A - B/(C + T)$$

Expansion type CPC



Schematic diagram of an automatic Nolan-Pollak CPC
O'Connor & McGovern. 1991

Expansion type CPC



Nolan-Pollak CPC

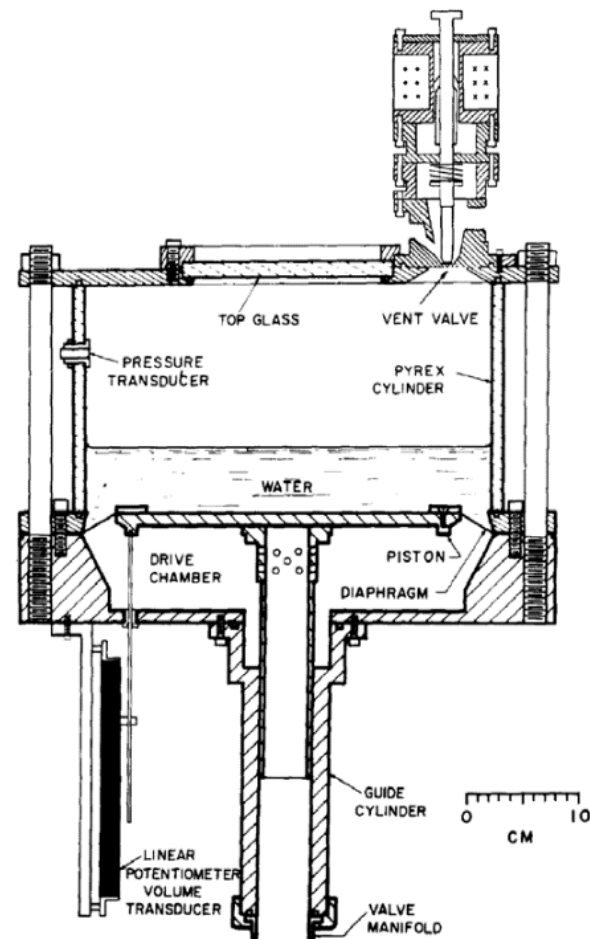


Fig. 3. A diagram of the UMR-AANC.

UMR-AANC (University of Missouri at Rolla-Absolute Aitken Nuclei Counter)

Continuous flow CPC

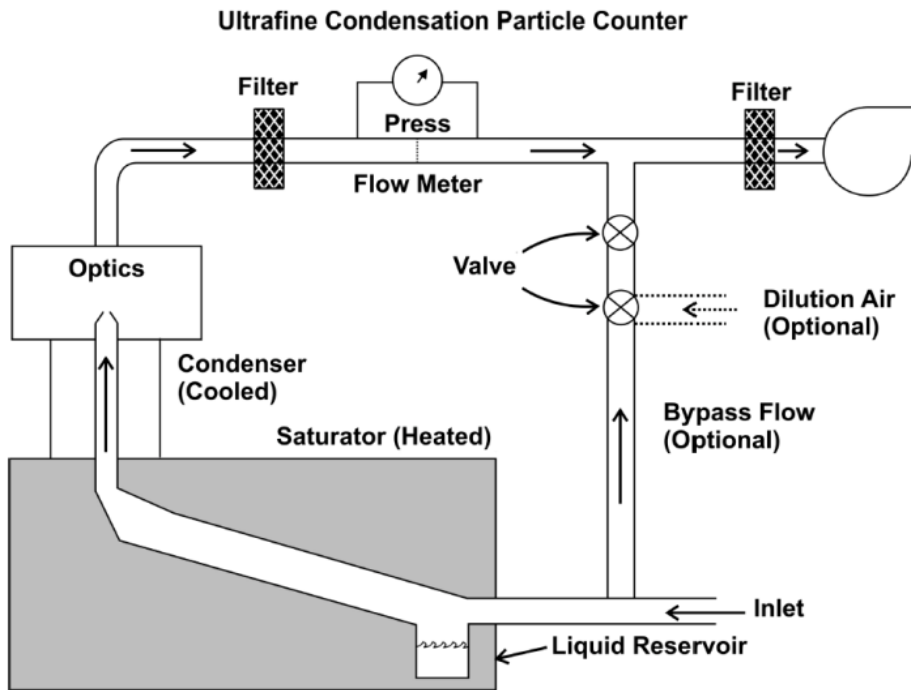
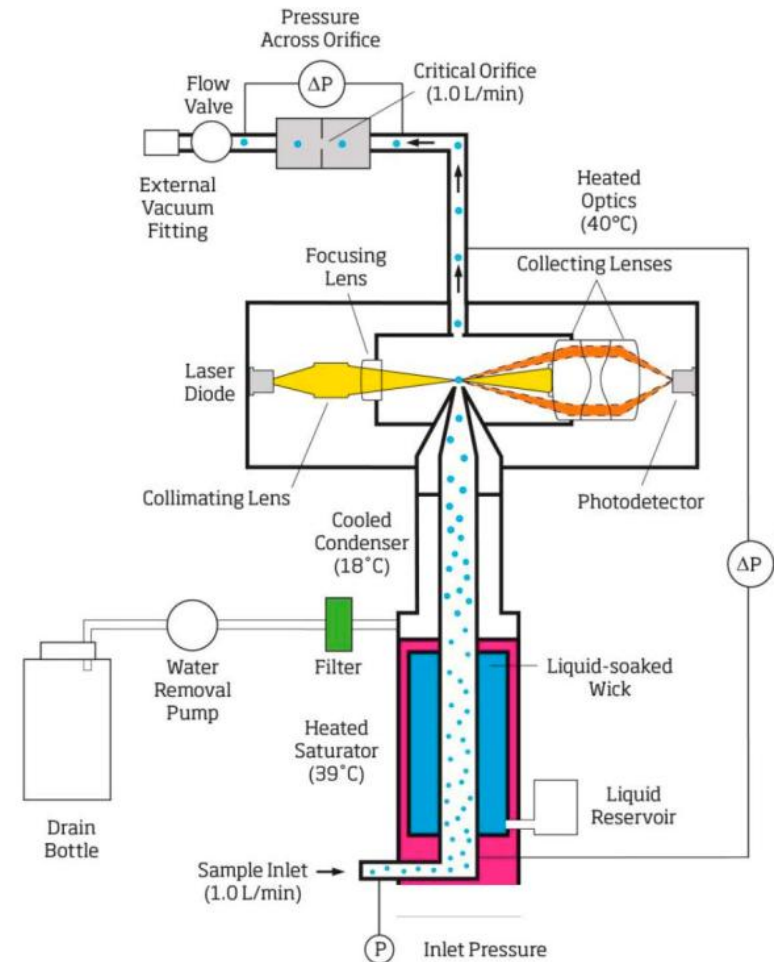


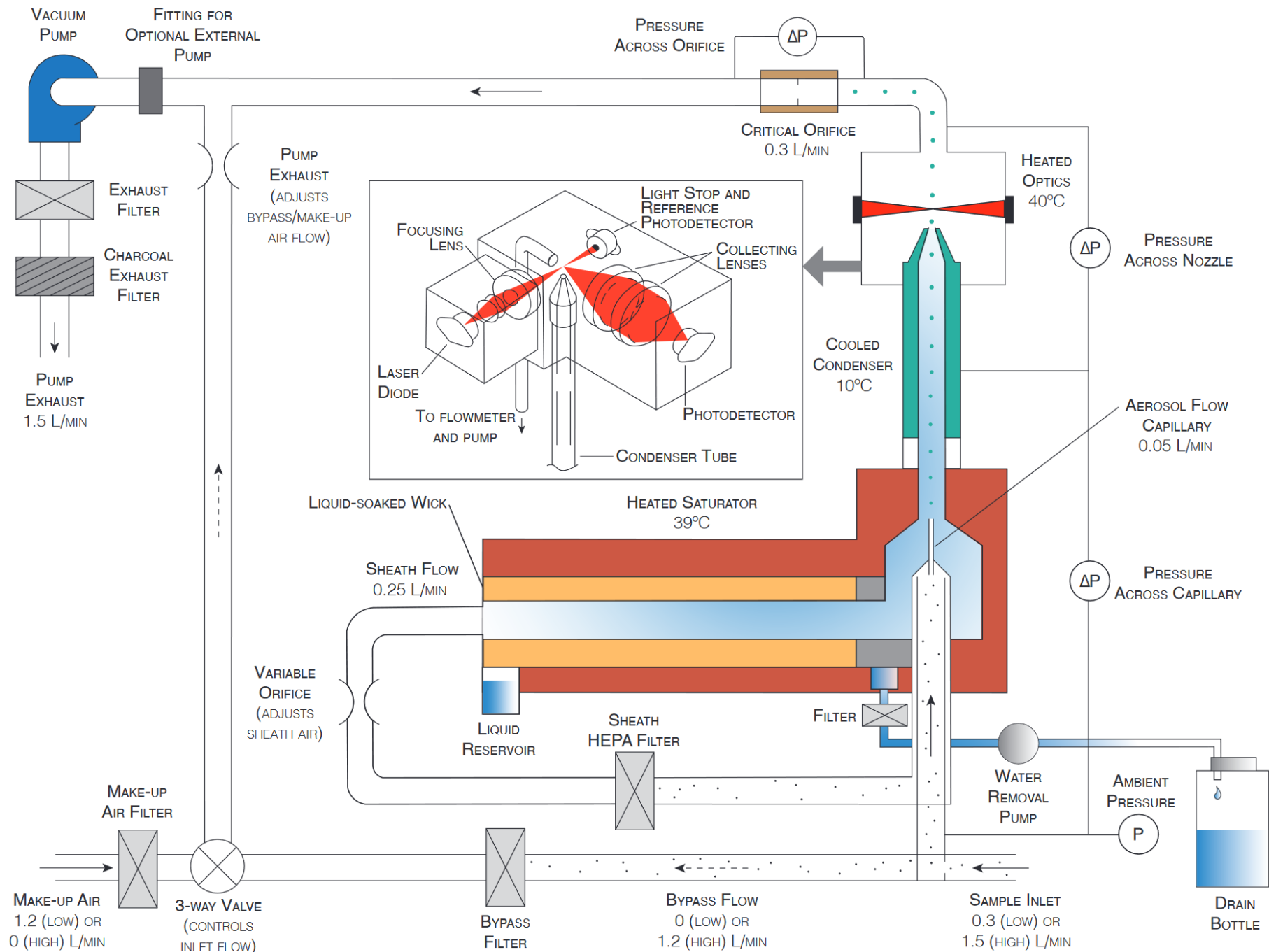
Figure 17-3 Schematic of a TSI model 3022 CPC.



TSI 3772 (3010, 3750)

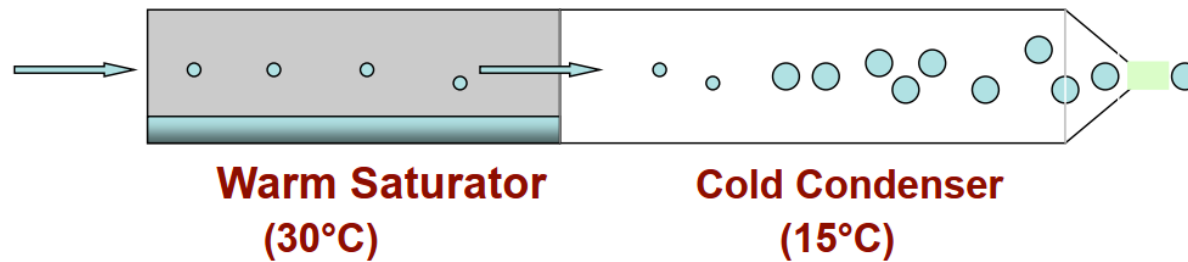
Expansion vs Continuous flow CPC

- Expansion CPC
 - Pro: uniform supersaturation
 - Con: cyclic, not continuous measurement
- Continuous CPC
 - Pro: continuous measurement
 - Con
 - Aerosols deposit onto the wall by diffusion
 - Higher supersaturation in flow center
 - Particles travels slow in flow center
 - Flammable working fluid
 - Can be partially alleviated via design

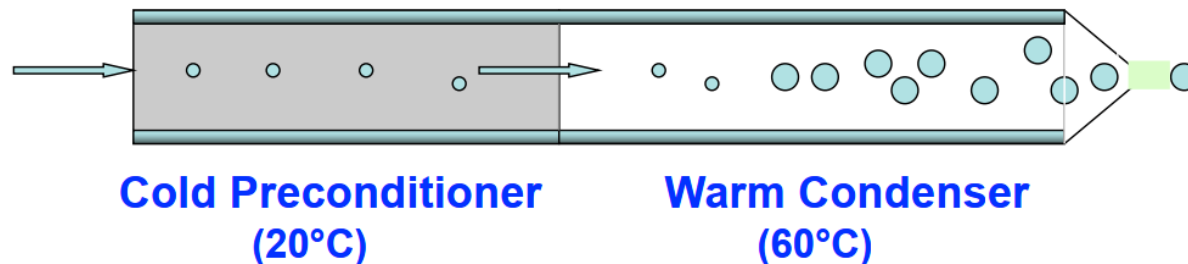


Water-based CPC

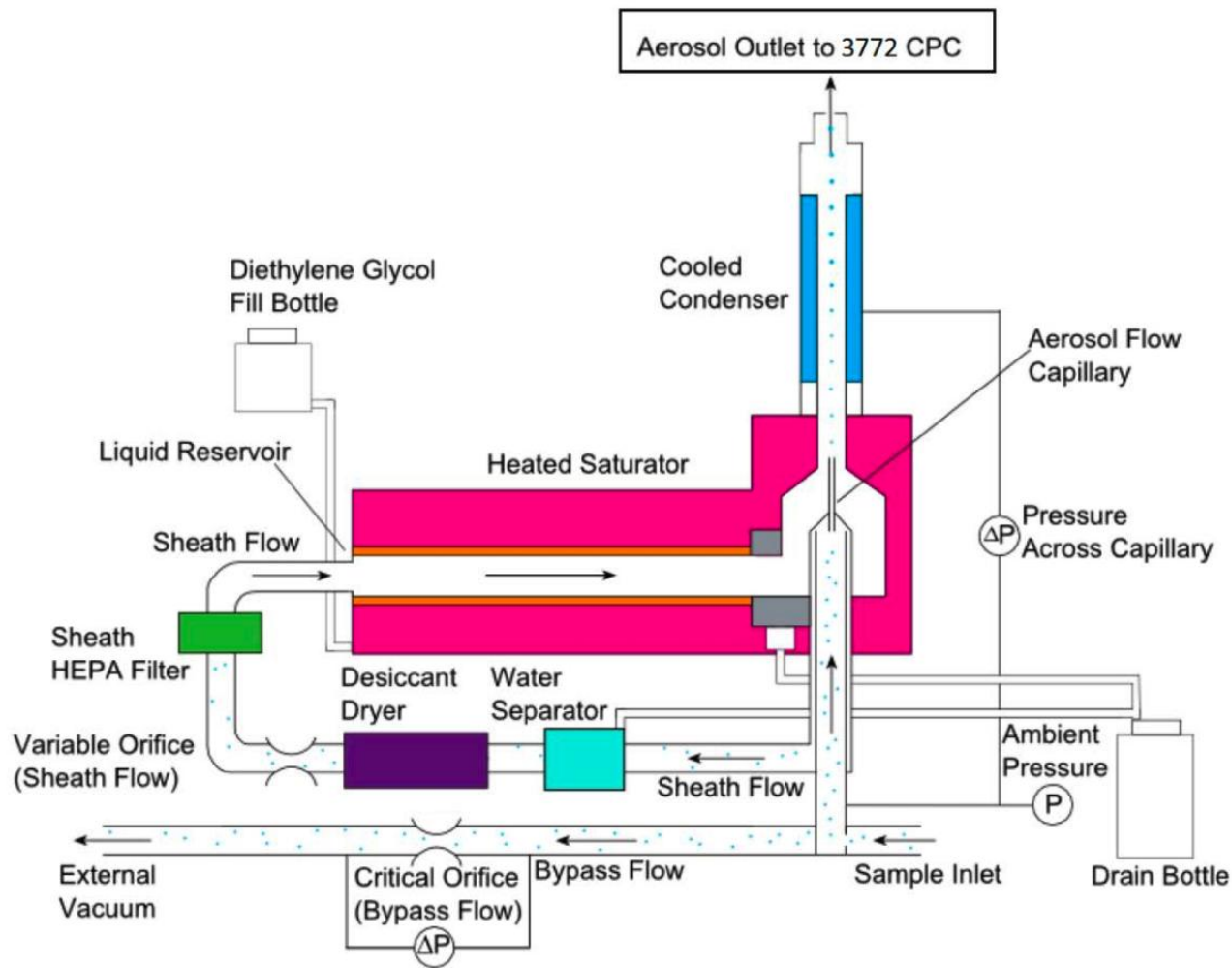
Alcohol-Based Continuous Flow :



Water-Based Continuous Flow :



Two-stage CPC



TSI 3777 nano enhancer (courtesy of TSI Inc.)

Applications and improvements for CPCs

- Currently widely used for
 - Studying new particle formation in the atmosphere
 - Monitoring of cleanrooms and indoor environments
 - Test of filters and respirators
 - Detection and size measurement of nanoparticles
- Limitations:
 - Response time
 - Minimum detection limit
 - Diffusion losses of small particles
 - Cost



TSI 3750 CPC



TSI 3789 WCPC